OPENKEYS R0.3 – ONE USB-C + NFC / WHITE 16x2 LCD
Fingerprint mandatory; PIN additional only
NOT a routed or qualified manufacturing release

01-usb



File: 01-usb.kicad_sch

02-power



File: 02-power.kicad_sch

03-mcu



File: 03-mcu.kicad_sch

04-se-storage



File: 04-se-storage.kicad_sch

05-nfc



File: 05-nfc.kicad_sch

06-trusted-ui



File: 06-trusted-ui.kicad_sch

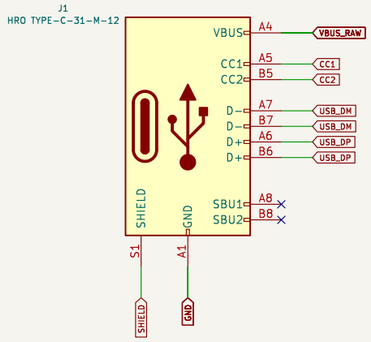
07-fault-tamper

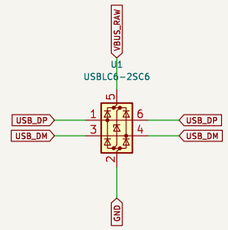


File: 07-fault-tamper.kicad_sch

Jay-Void-BA
Sheet: /
File: OpenKeys.kicad_sch
Title: OpenKeys R0.3 – Engineering review
Size: A3 | Date: 2026-11-08 | **Rev: 0.3**
KiCad E.D.A. 9.0.2 | Id: 1/8

One USB-C receptacle — USB 2.0 FS data and 5 V charging

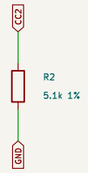


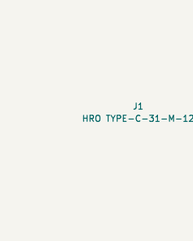






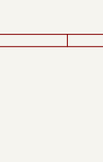


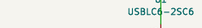












5.1k Rd on each CC pin: sink only. No PD negotiation, no 9 V request.
Apple 20 W adapter uses 5 V through a standards-compliant USB-C cable.
100mA hardware input limit: 5 V x 0.1 A = 0.50 W nominal input budget.
ESD/OVP coordination, inrush, CC leakage and USB suspend need EVT tests.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0.3 — 01-usb

Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 2/8

Low-rate charging, protected 500 mAh pack and 3.3 V supply

#FLG3
PWR_FLAG
USB_5V

VBUS_RAW
F1
250mA hold – qualify I2t
USB_5V

U2
BQ25185DLHR
SYS 1
BAT 2
CHG_STAT2_N 3
CHARGE_DISABLE 4
GND 5
BAT_NTC 6
SYS
BAT
STAT2
CE_N
GND
TS_MR
ILIM_VSET 7
ISET 8
STAT1 9
IN 10
EP 11
CHG_ILIM_VSET
CHG_ISET
CHG_STAT1_N
USB_5V
GND

J2
Protected 1S 500mAh + 10k NTC
BAT 1
BAT_NTC 2
GND 3

CHG_ILIM_VSET
R6
24k 1%
GND

CHG_ISET
R7
3.00k 1%
GND

SYS
R8
100k
CHARGE_DISABLE

CHARGE_DISABLE
CHARGE_ENABLE 1
Q2
2N7002
2
3
GND

CHARGE_ENABLE
R9
100k
GND

SW_L1
L1
1.5uH Isat>=1.5A
SW_L2

SYS
R10
0R
REG_VINA

#FLG4
PWR_FLAG
REG_VINA

U3
TPS63031DSK (3.3 V)
SYS 5
REG_VINA 8
REG_EN 6
GND 7
SW_L1 4
SW_L2 2
GND 9
GND 3
VIN
VINA
EN
PS/SYNC
L1
L2
GND
PGND
VOUT 1
FB 10
+3V3
+3V3

REG_EN 2
SW1
OFF / ON
1
3
SYS
GND

USB_5V
C2
1uF 10V
GND

BAT
C3
10uF 6.3V
GND

SYS
C4
10uF 10V
GND

SYS
C5
10uF 10V
GND

REG_VINA
C6
100nF
GND

+3V3
C7
10uF 6.3V
GND

+3V3
C8
10uF 6.3V
GND

Fixed100mA input; charge target100mA=0.2C for500mAh.
Input DPM gives system load priority, so cell charge can be slower.
CE defaults disabled. Public charging service never unlocks secrets.
Suspend: stop charge/backlight; qualify total VBUS current <=2.5mA.
Real pack PCM + attached thermistor required. No fast charge.

NOT FOR FABRICATION – see hardware/RELEASE_GATES.md

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0.3 – 02-power

Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 3/8

STM32H573RIT6 — STiRoT and TrustZone protected I/O

C9
100nF

C10
100nF

C11
100nF

C12
100nF

C13
100nF

C14
4.7uF

C15
2.2uF low ESR

C16
2.2uF low ESR

C17
100nF

C18
1uF

U4
STM32H573RIT6

FB1
600R@100MHz

#FLG6
PWR_FLAG

R11
10k

C19
10nF — qualify reset timing

R12
10k

C20
9pF C0G — tune

C21
9pF C0G — tune

Y1
ABS07-32768KHZ-6-T

J3
Internal SWD — DNP production

STiRoT must verify protected firmware. TrustZone/GTZC must deny nonsecure access to FP, SE, LCD, buttons, timers, DMA and key memory.
No raw cryptography or arbitrary display commands at secure gateway.
SE_EN or a sensor MATCH wire is NOT fingerprint authentication.
Separate2.2uF on VCAP1/2; no external load on VCAP.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

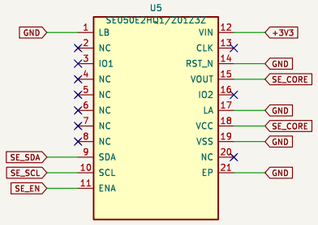
Jay-Void-BA

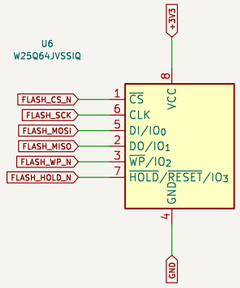
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

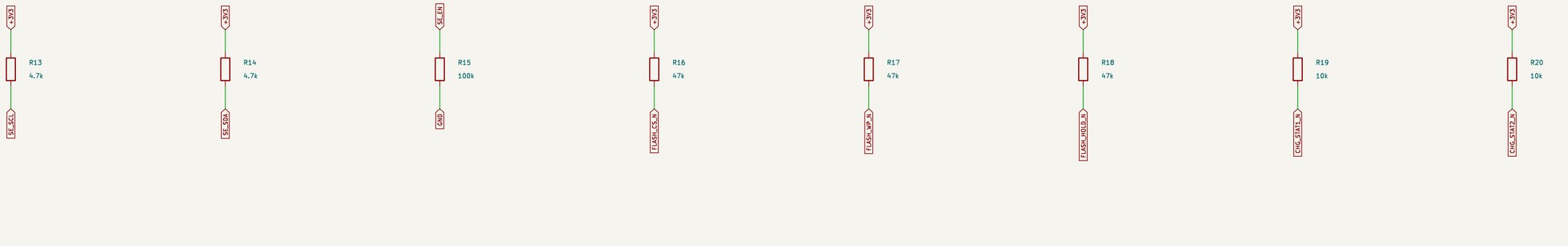
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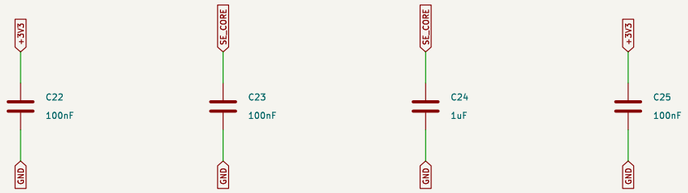
Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 4/8

Secure element and encrypted storage — no OTP / wall-clock RTC









Each user security operation: fresh authenticated device FP + confirmation.
Sensitive administration also needs device PIN. PIN never replaces FP.
Precharge attempts durably: uncertain writes deny. AEAD does not alone prevent rollback.
SE certification does not certify MCU, PCB, biometric path or complete product.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

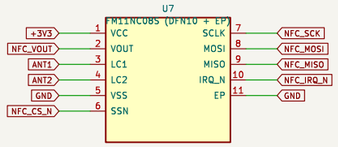
Jay-Void-BA

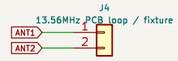
Sheet: /04-se-storage/
File: 04-se-storage.kicad_sch

Title: OpenKeys R0.3 — 04-se-storage

Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 5/8

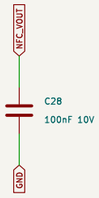
NFC Type4 card interface — battery-powered trusted fingerprint UI





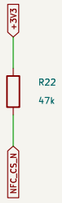












NFC VOUT isolated from SYS/BAT/+3V3. No assumed RF-powered full device.
FP/LCD/CTAP NFC require charged battery and device ON.
RF region: tuned coil only, no cell, metal shield or ground planes.
Mobile management needs authenticated APDUs and native OS integration.
Phone biometrics or a tap never replace device fingerprint verification.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

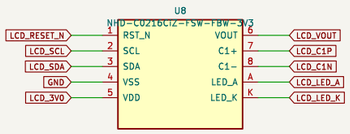
Jay-Void-BA

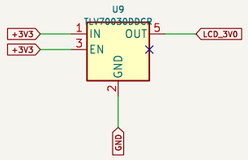
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File: 05-nfc.kicad_sch

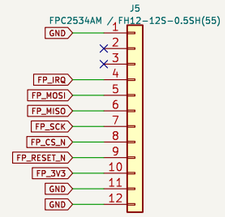
Title: OpenKeys R0.3 — 05-nfc

Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 6/8

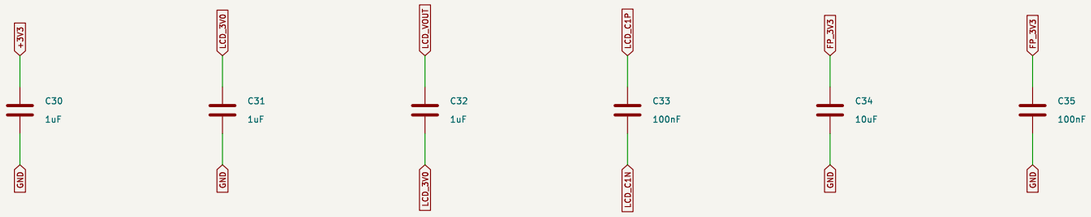
White-backlit16x2 LCD, authenticated fingerprint and contextual buttons

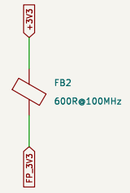


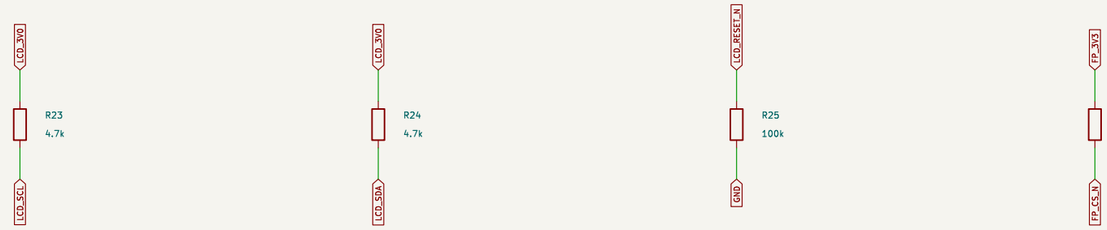


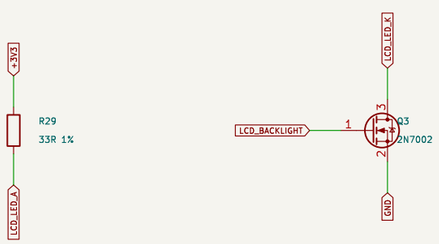
















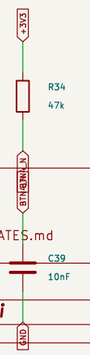












ENGLISH16x2; page full identity before confirmation.
PIN: short1..4=symbol; hold4=erase; hold2=cancel.
Menu:1=prev,2=next,3=confirm,4=cancel. No host injection.
Backlight about10mA via33R; default OFF.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

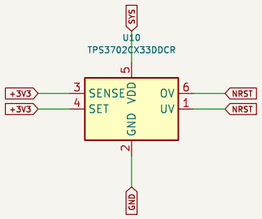
Jay-Void-BA

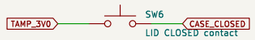
Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

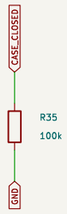
Title: OpenKeys R0.3 — 06-trusted-ui

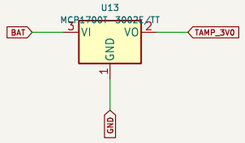
Size: A3	Date: 2026-11-08	**Rev: 0.3-design-review**
KiCad E.D.A. 9.0.2		Id: 7/8

Independent reset, battery–backed case tamper and SE interlock



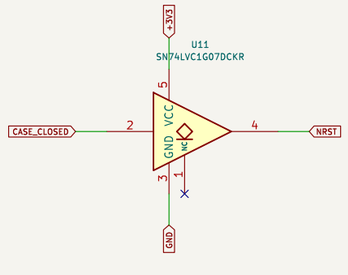


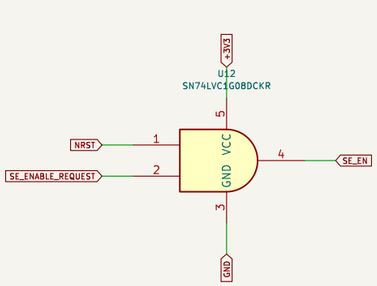




















Reset invalidates volatile grants; reboot must verify firmware and require fresh FP.
PC13=case input; PC1/TAMP_IN3 observes reset/fault line.
Internal BOR/PVD, clock checks and persistent precharged attempts also required.
Production debug lifecycle closed; no populated external debug connector.
VBAT+LSE must retain tamper with main OFF; full backup loss must lock, not initialize.
Measure reset/ENA latency, leakage and injection before security release.

NOT FOR FABRICATION – see hardware/RELEASE_GATES.md

Jay–Void–BA

Sheet: /07–fault–tamper/
File: 07–fault–tamper.kicad_sch

Title: OpenKeys R0.3 – 07–fault–tamper

Size: A3	Date: 2026–11–08	**Rev: 0.3–design–review**
KiCad E.D.A. 9.0.2		Id: 8/8